KAIZEN MATHS

# Lesson starter

**Algebraic substitution**

Powers and products

4 questions · about 5 minutes

Name _____

Evaluate each expression. Use brackets for negative values.

**S1** Given $a = 0, b = 0$

$$a^2 - 4ab + b^2$$

**S2** Given $a = 1, b = 0$

$$a^2 - 4ab + b^2$$

**S3** Given $a = 0, b = 2$

$$a^2 - 6ab + b^2$$

**S4** Given $a = 1, b = 4$

$$a^2 - 2ab + b^2$$



Show your working

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AS-05 · S · Starter

Show your working

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# Exit ticket

**Algebraic substitution**

Powers and products

Name _____

Evaluate each expression.

**E1** Given $a = 0, b = 2$

$$a^2 - 6ab + b^2$$

**E2** Given $a = 1, b = 4$

$$a^2 - 2ab + b^2$$

**E3** Given $a = 4, b = -1$

$$a^2 - 5ab + b^2$$

AS-05 · E · 3 questions

**Show your working**

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AS-05 · E · 3 questions

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# Teacher answers & working

AS-05 · Powers and products

Page 1: two starters. Page 2: four exit tickets. Cut along the dashed lines.

Print student pages single-sided at 100%. Match S and E labels to the solutions below.

## S1

Given a=0, b=0

$$a^2 - 4ab + b^2$$

Substitute a=0, b=0.

$$(0)^2 - 4(0)(0) + (0)^2$$

$$0$$

$$0$$

## S2

Given a=1, b=0

$$a^2 - 4ab + b^2$$

Substitute a=1, b=0.

$$(1)^2 - 4(1)(0) + (0)^2$$

$$1$$

$$1$$

## S3 / E1

Given a=0, b=2

$$a^2 - 6ab + b^2$$

Substitute a=0, b=2.

$$(0)^2 - 6(0)(2) + (2)^2$$

$$0 + 4$$

$$4$$

## S4 / E2

Given a=1, b=4

$$a^2 - 2ab + b^2$$

Substitute a=1, b=4.

$$(1)^2 - 2(1)(4) + (4)^2$$

$$1 - 8 + 16$$

$$9$$

## E3

Given a=4, b=-1

$$a^2 - 5ab + b^2$$

Substitute a=4, b=-1.

$$(4)^2 - 5(4)(-1) + (-1)^2$$

$$16 + 20 + 1$$

$$37$$